Exercise J-M

1. Let a function $f: R \to R$ be defined as

$$f(x)=\begin{cases}\sin x - e^x & \text{if } x \le 0\\ a+[-x] & \text{if } 0 < x < 1\\ 2x-b & \text{if } x \ge 1\end{cases}$$

where [x] is the greatest integer less than or equal to x. If *f* is continuous on R, then (a + b) is equal to : **[JEE(Main)July2021]**

माना एक फलन $f: R \to R$

$$f(x)=\begin{cases}\sin x - e^x & \text{यदि } x \le 0\\ a+[-x] & \text{यदि } 0 < x < 1\\ 2x-b & \text{यदि } x \ge 1\end{cases}$$

द्वारा परिभाषित है, जहाँ [x] महत्तम पूर्णांक ≤ x है। यदि फलन *f*, R पर संतत है, तो (a + b) बराबर है : **[JEE(Main)July2021]**

(A) 2 (B) 5 (C) 4 (D) 3

Ans. D

Sol. (i) Continuous at x = 0

LHL = - 1

f(0) = -1

RHL = a - 1

a - 1 = - 1

$\Rightarrow$ a = 0

(ii) Continuous at x = 1

LHL = a – 1

f(1) = 2 – b

RHL = 2 – b

a - 1 = 2 – b

$\Rightarrow$ b = 3

2. Let $f: R \to R$ be defined as $f(x)=\begin{cases}\dfrac{x^3}{(1-\cos 2x)^2}\log_e\left(\dfrac{1+2xe^{-2x}}{(1-xe^{-x})^2}\right) & , x \ne 0.\\ \alpha & , x = 0\end{cases}$

If f is continuous at x = 0, then α is equal to : **[JEE(Main)July2021]**

माना $f: R \to R$ $f(x) = \begin{cases} \dfrac{x^3}{(1-\cos 2x)^2} \log_e\left(\dfrac{1+2xe^{-2x}}{(1-xe^{-x})^2}\right) & , x \neq 0. \\ \alpha & , x = 0 \end{cases}$

द्वारा परिभाषित है। यदि $x = 0$ पर f संतत है, तो α बराबर है : **[JEE(Main)July2021]**

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. If function is continuous at $x = 0$

RHL = f(0)

$$\lim_{h\to 0} \frac{h^3}{(1-\cos 2h)^2} \ln\left(\frac{1+2he^{-h}}{(1-he^{-h})^2}\right) = \alpha$$

$$\lim_{h\to 0} \frac{h^3}{4\sin^4 h}\left[\ln(1+2he^{-h}) - 2\ln(1-he^{-h})\right] = \alpha$$

$$\lim_{h\to 0} \frac{1}{4h}\left[\frac{\ln(1+2he^{-2h})}{2he^{-2h}}\left(2he^{-2h}\right) - \frac{2\ln(1-he^{-h})}{-he^{-h}}\left(-he^{-h}\right)\right] = \alpha$$

$$\lim_{h\to 0} \frac{1}{4h}\left[2he^{-2h} + 2he^{-h}\right] = \alpha$$

$$\lim_{h\to 0} \frac{2}{4}\left(e^{-2h} + e^{-h}\right) = \alpha$$

$\alpha = 1$

3. Let $f: R \to R$ be defined as

$$f(x) = \begin{cases} \dfrac{\lambda|x^2-5x+6|}{\mu(5x-x^2-6)}, & x < 2 \\ e^{\frac{\tan(x-2)}{x-[x]}} & , x > 2 \\ \mu & , x = 2 \end{cases}$$

where [x] is the greatest integer less than or equal to x. If f is continuous at $x = 2$, then $\lambda + \mu$:

[JEE(Main)July2021]

माना $f: R \to R$

$$f(x) = \begin{cases} \dfrac{\lambda\,|\,x^2-5x+6\,|}{\mu(5x-x^2-6)}, & x < 2 \\ e^{\frac{\tan(x-2)}{x-[x]}}, & x > 2 \\ \mu, & x = 2 \end{cases}$$

द्वारा परिभाषित है, जबकि [x] महत्तम पूर्णांक ≤ x है। यदि x = 2 पर f संतत है, तो λ + μ बराबर है :

[JEE(Main)July2021]

(A) 1 (B) 2e – 1 (C) e(e – 1) (D) e(–e + 1)

Ans. D

Sol. $f(x)$ is continuous at $x = 2$

$f(2^-) = f(2) = f(2^+)$

$$\lim_{x\to 2^-} \frac{\lambda\,|\,x^2-5x+6\,|}{-(x^2-5x+6)\mu} = \mu = \lim_{x\to 2^+} e^{\frac{\tan(x-2)}{x-[x]}}$$

$$-\frac{\lambda}{\mu} = \mu = e$$

$\mu = e, \quad \lambda = -e^2$

$\lambda + \mu = e(-e + 1)$

4. Let $f: \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \to R$ be defined as

$$f(x) = \begin{cases} (1+|\sin x|)^{\frac{3a}{|\sin x|}}, & -\frac{\pi}{4} < x < 0 \\ b, & x = 0 \\ e^{\cot 4x/\cot 2x}, & 0 < x < \frac{\pi}{4} \end{cases}.$$

If f is continuous at $x = 0$, then the value of $6a + b^2$ is equal to : **[JEE(Main)July2021]**

माना $f: \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \to R$,

$$f(x) = \begin{cases} (1+|\sin x|)^{\frac{3a}{|\sin x|}} & , \quad -\frac{\pi}{4} < x < 0 \\ b & , \quad x = 0 \\ e^{\cot 4x / \cot 2x} & , \quad 0 < x < \frac{\pi}{4} \end{cases}$$

द्वारा परिभाषित है। यदि $x = 0$ पर f संतत है, $6a + b^2$ का मान बराबर है : **[JEE(Main)July2021]**

(A) $e - 1$ (B) $1 - e$ (C) e (D) $1 + e$

Ans. D

Sol. $f(0) = f(0^+) = f(0^-)$

$$b = \lim_{x \to 0^+} (1 + |\sin x|)^{\frac{3a}{|\sin x|}} = \lim_{x \to 0^-} e^{\frac{\cot 4x}{\cot 2x}}$$

$$b = e^{\lim_{x \to 0^+} \frac{3a}{|\sin x|}(1+|\sin x|-1)} = \lim_{x \to 0^-} e^{\frac{\tan 2x}{\tan 4x}}$$

$$b = e^{3a} = e^{1/2} \Rightarrow a = \frac{1}{6}, \; b = \sqrt{e}$$

$6a + b^2 = 1 + e$

5. Let $f: [0, \infty) \to [0, 3]$ be a function defined by $f(x) = \begin{cases} \max\{\sin t : 0 \le t \le x\}, & 0 \le x \le \pi \\ 2 + \cos x, & x > \pi \end{cases}$ Then which of the following is true : **[JEE(Main)July2021]**

(A) f is not continuous exactly at two points in $(0, \infty)$

(B) f is differentiable everywhere in $(0, \infty)$

(C) f is continuous everywhere but not differentiable exactly at one point in $(0, \infty)$

(D) f is continuous everywhere but not differentiable exactly at two points in $(0, \infty)$

माना $f: [0, \infty) \to [0, 3]$ $f(x) = \begin{cases} \max\{\sin t : 0 \le t \le x\}, & 0 \le x \le \pi \\ 2 + \cos x, & x > \pi \end{cases}$ द्वारा परिभाषित है। तो निम्न में से कौनसा सत्य है : **[JEE(Main)July2021]**

(A) $(0, \infty)$ के ठीक दो बिन्दु पर f संतत नहीं है

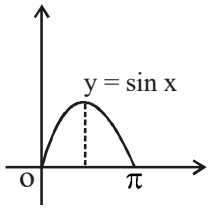
(B) $(0, \infty)$के प्रत्येक बिन्दु पर f अवकलनीय है

(C) $(0, \infty)$के प्रत्येक बिन्दु पर f संतत है परन्तु ठीक एक बिन्दु अवकलनीय नहीं है

(D) $(0, \infty)$के प्रत्येक बिन्दु पर f संतत है परन्तु ठीक दो बिन्दुओं पर अवकलनीय नहीं है

Ans. B

Sol.



$$f(x)=\begin{cases}\sin x & 0\le x<\dfrac{\pi}{2}\\ 1 & \dfrac{\pi}{2}\le x\le\pi\\ 2+\cos x & x>\pi\end{cases}$$

Doubtful points: $x=\dfrac{\pi}{2},\pi$

$\text{LHL}=\text{RHL}=f\left(\dfrac{\pi}{2}\right)=1 \quad \therefore$ hence Continuous at $x=\dfrac{\pi}{2}$

$\text{LHL}=\text{RHL}=f(\pi)=1 \quad \therefore$ hence Continuous at $x=\pi$

$$f'(x)=\begin{cases}\cos x & 0<x<\dfrac{\pi}{2}\\ 0 & \dfrac{\pi}{2}<x<\pi\\ -\sin x & x>\pi\end{cases}$$

$f'\left(\dfrac{\pi^-}{2}\right)=F'\left(\dfrac{\pi^+}{2}\right)=o \quad \therefore$ Derivable at $x=\dfrac{\pi}{2}$

$f'(\pi^-)=f'(\pi^+)=o \quad \therefore$ Derivable at $x=\pi$

$\therefore$ f is differentiable everywhere in $(0,\infty)$

6. Let a function $g:[0,4]\to R$ be defined as $g(x)=\begin{cases}\max\{t^3-6t^2+9t-3\}, 0\le x\le 3, 0\le t\le x\\ 4-x \qquad\qquad , 3<x\le 4\end{cases}$, then the number of points in the interval (0, 4) where g(x) is NOT differentiable, is _____: **[JEE(Main)July2021]**

माना एक फलन $g : [0, 4] \to R$, $g(x) = \begin{cases} \max\{t^3 - 6t^2 + 9t - 3\}, 0 \le x \le 3, 0 \le t \le x \\ 4 - x \qquad\qquad , 3 < x \le 4 \end{cases}$ द्वारा परिभाषित है, तो अन्तराल

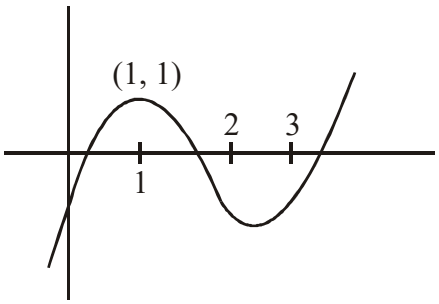
(0, 4) में जिन बिन्दुओं पर g(x) अवकलनीय नहीं है, उनकी संख्या है _____: **[JEE(Main)July2021]**

Ans. 1

Sol. $y = t^3 - 6t^2 + 9t - 3$

$$\frac{dy}{dx} = 3t^2 - 12t + 9 = 0$$

$t = 1 \qquad t = 3$



$g(x) \to x^3 - 6x^2 + 9x - 3 \qquad 0 \le n \le 1$

$\to 1 \qquad 1 < x \le 3$

$\to 4 - x \qquad 3 < x \le 4$

$g'(x) \to 3x^2 - 12x + 9 \qquad 0 \le x \le 1$

$\to 0 \qquad 1 < n \le 3$

$\to 4 \qquad 3 < x \le 4$

$g'(1^-) = g'(1^+)$ but $g'(3^-) \neq g'(3^+)$

nondifferentiable at one point

7. Let $f : R \to R$ be a function defined as $f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & \text{if } |x| \le 2 \\ 0 & \text{if } |x| > 2 \end{cases}$. Let $g : R \to R$ be given by

$g(x) = f(x + 2) - f(x - 2)$. If n and m denote the number of points in R where g is not continuous and not differentiable, respectively, then n + m is equal to _____. **[JEE(Main)July2021]**

माना फलन $f : R \to R$

$$f(x) = \begin{cases} 3\left(1-\frac{|x|}{2}\right) & ; \text{जब} |x| \le 2 \text{ हो} \\ 0 & ; \text{जब} |x| > 2 \text{ हो} \end{cases}$$

द्वारा परिभाषित है। माना $g : R \to R$ $g(x) = f(x + 2) – f(x –2)$ द्वारा परिभाषित है। यदि R के उन बिन्दुओं की संख्या जहाँ g संतत नहीं है और जहाँ g अवकलनीय नहीं है, क्रमशः n और m हैं, तो $n + m$ बराबर है _____

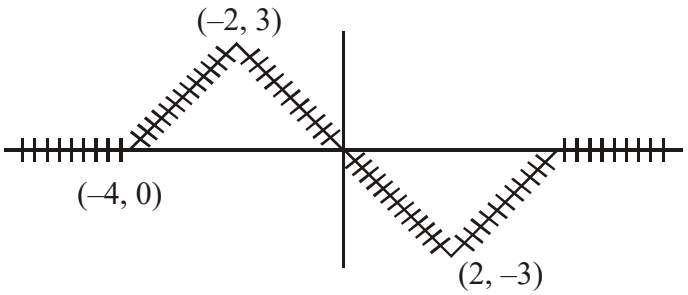
[JEE(Main)July2021]

Ans. 4

Sol. $$f(x–2) = \begin{cases} \frac{3x}{2} & : \quad -4 \le x \le -2 \\ \frac{-3x}{2} & : \quad -2 < x \le 0 \\ 0 & : \quad x \in (-\infty,-4) \cup (0,\infty) \end{cases}$$

$$f(x–2) = \begin{cases} \frac{3x}{2} & : \quad 0 \le x \le 2 \\ \frac{-3x}{2}+6 & : \quad 2 \le x \le 4 \\ 0 & : \quad x \in (-\infty,0) \cup (4,\infty) \end{cases}$$

$$g(x) = f(x + 2) – f(x – 2) = \begin{cases} \frac{3x}{2}+6 & : \quad -4 \le x \le -2 \\ \frac{-3x}{2} & : \quad -2 < x < 2 \\ \frac{3x}{2}-6 & : \quad 2 \le x \le 4 \\ 0 & : \quad x \in (-\infty,-4) \cup (4,\infty) \end{cases}$$



here $n = 0$

$m = 4$

hence $m + n = 4$

8. Consider the function $f(x) = \frac{P(x)}{\sin(x-2)}, \quad x \neq 2$

$= 7 \qquad , x = 2$

where P(x) is a polynomial such that P''(x) is always a constant and P(3) = 9. If f(x) is continuous at x = 2, then P(5) is equal to _____. **[JEE(Main)July2021]**

फलन $f(x) = \frac{P(x)}{\sin(x-2)}, \quad x \neq 2$

$= 7, \qquad x = 2$

का विचार कीजिए, जहाँ P(x) एक बहुपद है, जिसके लिए P''(x) हमेशा एक अचर है तथा P(3) = 9 है। यदि x = 2 पर f(x) संतत हैं, तो P(5) बराबर है _____ **[JEE(Main)July2021]**

Ans. 39

Sol. $f(x) = \begin{cases} \frac{P(x)}{\sin(x-2)} & x \neq 2 \\ 7 & x = 2 \end{cases}$

P'' (x) = constant $\Rightarrow$ P(x) is 2 degree polynomial

$f(2^+) = f(2^-) = f(2)$

$\lim\limits_{x\to 2} \frac{P(x)}{\sin(x-2)} = 7$

$\lim\limits_{x\to 2} \frac{(x-2)P(x)}{(x-2)\sin(x-2)} = 7$

$\lim\limits_{x\to 2} \frac{P(x)}{(x-2)} = 7$

$P(x) = (x-2)(ax+b)$

$2a + b = 7$

$P(3) = 3a + b = 9$

$a = 2$

$4 + b = 7 \Rightarrow b = 3$

$P(x) = (x-2)(2x+3)$

$P(5) = 3(10 + 3) = 39$

9. Let f: [0,3] $\to$ R be defined by

$f(x) = \min \{x - [x], 1 + [x] - x\}$

where [x] is the greatest integer less than or equal to x.

Let P denote the set containing all $x \in [0,3]$ where *f* is discontinuous, and Q denote the set containing all $x \in (0,3)$ where *f* is not differentiable. Then the sum of number of elements in P and Q is equal to_____:

[JEE(Main)July2021]

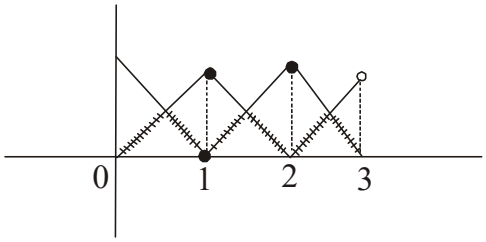
माना f: [0,3] $\to$ R,

$f(x) = \min \{x - [x], 1 + [x] - x\}$, जहाँ [x] महत्तम पूर्णांक $\le x$ है, द्वारा परिभाषित है। माना सभी $x \in [0,3]$ जिनके लिए f असंतत है, का समुच्चय P है तथा सभी $x \in (0,3)$, जिनके लिए f अवकलनीय नहीं है का समुच्चय Q है। तो P तथा Q में अवयवों की संख्या का योगफल बराबर है _____

[JEE(Main)July2021]

Ans. 5

Sol. $f(x) = \min\left(\{x\}, 1 - \{x\}\right)$



n(P) = 0

n(Q) = 5

n (P) + n (Q) = 5

10. Let [t] denote the greatest integer less than or equal to t. Let f(x) = x – [x], g(x) = 1 – x + [x], and h(x) = min {f(x), g(x)}. $x \in [-2, 2]$. Then h is : **[JEE(Main)Aug.2021]**

(A) Continuous in [– 2, 2] but not differentiable at exactly three points in (– 2, 2)

(B) Not continuous at exactly four points in [–1, 2]

(C) Not continuous at exactly three points in [–1, 2]

(D) Continuous in [– 2, 2] but not differentiable at more than four points in (–2, 2)

माना [t] महत्तम पूर्णांक $\le$ t है। माना f(x) = x – [x], g(x) = 1 – x + [x] तथा h(x) = min {f(x), g(x)}, $x \in [-2, 2]$ है, तो h :

[JEE(Main)Aug.2021]

(A) [– 2, 2] में संतत है परन्तु (– 2, 2) में ठीक तीन बिन्दुओं से अधिक पर अवकलनीय नहीं है

(B) [–1, 2] में ठीक तीन बिन्दुओं पर संतत नहीं है।

(C) [–1, 2] में ठीक चार बिन्दुओं पर संतत नहीं है।

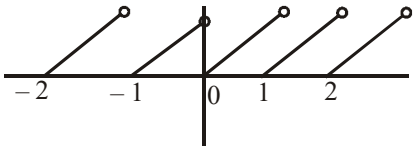
(D) [– 2, 2] में संतत है परन्तु (–2, 2) में चार बिन्दुओं से अधिक पर अवकलनीय नहीं है।

Ans. D

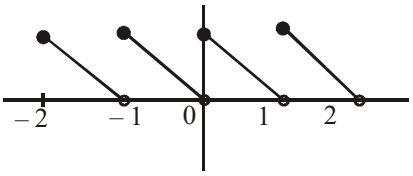
Sol. $f(x) = \{x\}$

$g(x) = 1 - \{x\}$

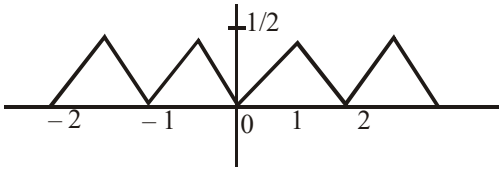
$y = \{x\}$



$g(x) = 1 - \{x\}$



$h(x) = \min\left(f(x), g(x)\right)$



non differentiable at 7 points

11. The function $f(x) = |x^2 - 2x - 3|.\ e^{|9x^2 - 12x + 4|}$ is not differentiable at exactly : **[JEE(Main)Aug.2021]**

(A) one point (B) three points (C) two points (C) four points

फलन $f(x) = |x^2 - 2x - 3|.\ e^{|9x^2 - 12x + 4|}$: **[JEE(Main)Aug.2021]**

(A) ठीक एक बिन्दु पर अवकलनीय नहीं है (2) ठीक तीन बिन्दुओं पर अवकलनीय नहीं है

(C) ठीक दो बिन्दुओं पर अवकलनीय नहीं है (D) ठीक चार बिन्दुओं पर अवकलनीय नहीं है

Ans. C

Sol. $f(x) = |x^2 - 2x - 3|\, e^{|(3x-2)^2|}$

$f(x) = |(x-3)(x+1)|\, e^{(3x-2)^2}$

Not differentiable at $x = -183$

12. If the function $f(x) = \begin{cases} \frac{1}{x}\log_e\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right) & , \quad x < 0 \\ k & , \quad x = 0 \\ \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2+1}-1} & , \quad x > 0 \end{cases}$ is continuous at $x = 0$, then $\frac{1}{a} + \frac{1}{b} + \frac{4}{k}$ is equal to :

[JEE(Main)Aug.2021]

यदि फलन $f(x)=\begin{cases}\frac{1}{x}\log_e\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right) & , \quad x<0 \\ k & , \quad x=0 \\ \frac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1} & , \quad x>0\end{cases}$

$x=0$ पर संतत हैं, तो $\frac{1}{a}+\frac{1}{b}+\frac{4}{k}$ बराबर है : **[JEE(Main)Aug.2021]**

(A) 5 (B) – 5 (C) – 4 (D) 4

Ans. B

Sol. $f(0)=k$

$$f(0^+)=\lim_{x\to 0^+}\frac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1}\times\frac{\sqrt{x^2+1}+1}{\sqrt{x^2+1}+1}$$

$$f(0^+)=\lim_{x\to 0^+}\frac{-2\sin^2 x}{x^2}(\sqrt{x^2+1}+1)$$

$$f(0^+)=-4$$

$$f(0^-)=\lim_{x\to 0^-}\frac{1}{x}\log_e\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right)$$

$(9\sqrt{2},2)$

$$f(0^+)=\frac{1}{a}+\frac{1}{b}$$

$$f(0)=f(0^+)=f(0^-)$$

$$k=-4=\frac{1}{a}+\frac{1}{b}\Rightarrow\frac{1}{a}+\frac{1}{b}+\frac{4}{k}=-5$$

13. Let $a,b\in R, b\neq 0$. Define a function

$$f(x)=\begin{cases}a\sin\frac{\pi}{2}(x-1), & \text{for } x\le 0 \\ \frac{\tan 2x-\sin 2x}{bx^3}, & \text{for } x>0.\end{cases}$$

If f is continuous at $x=0$, then $10-ab$ is equal to _____. **[JEE(Main)Aug.2021]**

माना $a, b \in R, b \neq 0$ है। एक फलन

$$f(x) = \begin{cases} a\sin\frac{\pi}{2}(x-1), & \text{के लिए } x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3}, & \text{के लिए } x > 0. \end{cases}$$

द्वारा परिभाषित है। यदि x = 0 पर f संतत है, तो 10 – ab बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 14

Sol. $f(0) = -a = LHL$

$$RHL = \lim_{x\to 0}\frac{\tan 2x - \sin 2x}{bx^3} = \frac{4}{b}$$

$$-a = \frac{4}{b} \Rightarrow ab = -4$$

$10 - ab = 10 + 4 = 14$

14. Let [t] denote the greatest integer ≤ t. The number of points where the function $f(x) = [x]|x^2 - 1| + \sin\left(\frac{\pi}{[x]+3}\right) - [x+1], x \in (-2,2)$ is not continuous is _____

[JEE(Main)Aug.2021]

माना [t] महत्तम पूर्णांक ≤ t है। जहाँ फलन $f(x) = [x]|x^2 - 1| + \sin\left(\frac{\pi}{[x]+3}\right) - [x+1], x \in (-2,2)$ संतत नहीं है, ऐसे बिन्दुओं की संख्या है _____ **[JEE(Main)Aug.2021]**

Ans. 2

Sol. $f(x) = [x]|x^2 - 1| + \sin\left(\frac{\pi}{[x]+3}\right) - [x+1]$

$x \in (-2,2)$

$$f(x) = [x]\left(|x^2 - 1| - 1\right) + \sin\left(\frac{\pi}{[x]+3}\right) - 1$$

$$f(x) = (-2)\left((x^2 - 1) - 1\right) + \sin\left(\frac{\pi}{1}\right) - 1; \ (-2,-1]$$

$$= (-1)(-x^2 + 1 - 1) + \sin\left(\frac{\pi}{2}\right) - 1; \ (-1,0)$$

$$= 0 + \sin\frac{\pi}{3} - 1 \quad ; \ (0,1)$$

$= (1)((x^2-1)-1) + \sin\frac{\pi}{4} - 1;\ (1,2)$

$$f(x)\begin{cases} -2(x^2-2)-1 & (-2,-1] \\ x^2 & (-1,0) \\ \sin\frac{\pi}{3}-1 & [0,1) \\ (x^2-3)+\sin\frac{\pi}{4} & [1,2) \end{cases}$$

discont. at $x = 0, 1$.

15. Let $f(x) = \begin{cases} \frac{\sin(x-[x])}{x-[x]} & , \quad x \in (-2,-1) \\ \max\{2x, 3[|x|]\} & , \quad |x| < 1 \\ 1 & , \quad \text{otherwise} \end{cases}$

Where [t] denotes greatest integer $\le$ t. If m is the number of points where f is not continuous and n is the number of points where f is not differentiable, then the ordered pair (m, n) is : **[JEE(Main)June2022]**

माना $f(x) = \begin{cases} \frac{\sin(x-[x])}{x-[x]} & , \quad x \in (-2,-1) \\ \max\{2x, 3[|x|]\} & , \quad |x| < 1 \\ 1 & , \quad \text{otherwise} \end{cases}$

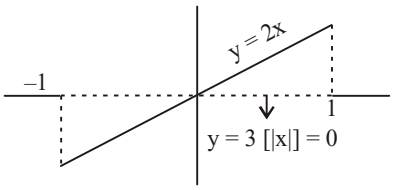
जहाँ [t] महत्तम पूर्णांक $\le$ t है। यदि m बिन्दुओं की संख्या है, जहाँ f संतत नहीं है, तथा n उन बिन्दुओं की संख्या है, जहाँ f अवकलनीय नहीं है, तो क्रमित युग्म (m, n) है : **[JEE(Main)June2022]**

(A) (3, 3) (B) (2, 4) (C) (2, 3) (D) (3, 4)

Ans. C

Sol. $f(x) = \begin{cases} \frac{\sin\{x\}}{\{x\}} & : \quad x \in (-2,-1) \\ \max(2x, 3[|x|]) & : \quad -1 < x < 1 \\ 1 & : \quad \text{otherwise} \end{cases}$

Let consider $y = \max(2x_1, 3[|x|]) : -1 , x < 1$



$$y = \max(2x, 0) = \begin{cases} 0 & : \ -1 < x < 0 \\ 2x & : \ 0 \le x < 1 \end{cases}$$

Now doubtful points for continuity and differentiability are $x = 0, 1, -1$

Now continuity at $x = 0$

RHL = 0

f(0) = 0

LHL = 0

Now RHD = 2

LHD = 0

Continuity at $x = 1$

LHL = 2

RHL = 1

f(1) = 1

Continuity at $x = -1$

RHL = 0

$$\text{LHL} = \frac{\sin(1-h)}{1-h} = \sin 1$$

f(–1) = 1

hence it is clear function is not continuous at x = 1, – 1 and not differentiable at x = 1, –1, 0

m = 2

n = 3

(2, 3)

16. Let f, g : R → R be two real valued functions defined as $f(x) = \begin{cases} -|x+3| & , \ x < 0 \\ e^x & , \ x \ge 0 \end{cases}$ and $g(x) = \begin{cases} x^2 + k_1 x & , \ x < 0 \\ 4x + k_2 & , \ x \ge 0 \end{cases}$, where k_1 and k_2 are real constants. If (gof) is differentiable at x = 0, then (gof) (–4) + (gof) (4) is equal to : **[JEE(Main)June2022]**

माना दो वास्तविक मान फलन $f, g : R \to R$

$$f(x) = \begin{cases} -|x+3| & , \ x < 0 \\ e^x & , \ x \ge 0 \end{cases} \text{ तथा } g(x) = \begin{cases} x^2 + k_1 x & , \ x < 0 \\ 4x + k_2 & , \ x \ge 0 \end{cases}$$

द्वारा परिभाषित हैं, जहाँ k_1 तथा k_2 वास्तविक अचर हैं। यदि x = 0 पर (gof) अवकलनीय है, तो (gof) (–4) + (gof) (4) बराबर है:

[JEE(Main)June2022]

(A) $4(e^4 + 1)$ (B) $2(2e^4 + 1)$ (C) $4e^4$ (D) $2(2e^4 - 1)$

Ans. D

Sol. $$f(x) = \begin{Bmatrix} x+3 & ; & x < -3 \\ -(x+3) & ; & -3 \le x < 0 \\ e^x & ; & x \ge 0 \end{Bmatrix}$$

$$g(x) = \begin{Bmatrix} x^2 + k_1 x & ; & x < 0 \\ 4x + k_2 & ; & x \ge 0 \end{Bmatrix}$$

$$g(f(x)) = \begin{Bmatrix} f(x)^2 + k_1 f(x) & ; & f(x) < 0 \\ 4f(x) + k_2 & ; & f(x) \ge 0 \end{Bmatrix}$$

$$g(f(x)) = \begin{Bmatrix} (x+3)^2 + k_1(x+3) & ; & x < -3 \\ (x+3)^2 - k_1(x+3) & ; & -3 \le x < 0 \\ 4e^x + k_2 & ; & x > 0 \end{Bmatrix}$$

check continuity at $x = 0$

$gof(0) = f(f(0^-)) = g(f(0^+))$

$4 + k_2 = 9 - 3k_1 = 4 + k_2$

$3k_1 + k_2 = 5$ ...(a)

differentiate

$$(g(f(x)))' = \begin{Bmatrix} 2(x+3) + k_1 & ; & x < -3 \\ 2(x+3) - k_1 & ; & -3 \le x < 0 \\ 4e^x & ; & x \ge 0 \end{Bmatrix}$$

$6 - k_1 = 4$

$k_1 = 2$ ...(b)

$\therefore k_1 = 2, k_2 = -1$

$$gof(x) = \begin{Bmatrix} (x+3)^2 + 2(x+3) & ; & x < -3 \\ (x+3)^2 - 2(x+3) & ; & -3 \le x < 0 \\ 4e^x - 1 & ; & x \ge 0 \end{Bmatrix}$$

$gof(-4) + gof(4) = 4e^4 - 2$

$\Rightarrow 2(2e^4 - 1)$

17. Let $f(x) = \min\{1, 1 + x \sin x\}$, $0 \le x \le 2\pi$. If m is the number of points, where f is not differentiable and n is the number of points, where f is not continuous, then the ordered pair (m, n) is equal to :

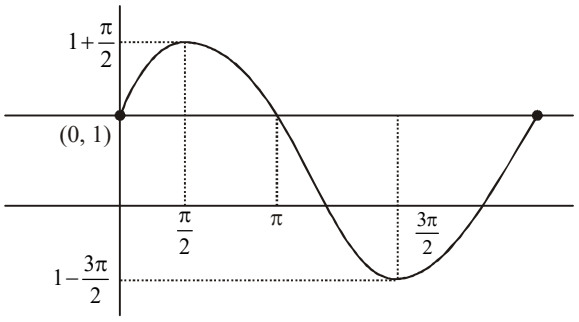
[JEE(Main)June2022]

माना $f(x) = \min\{1, 1 + x\sin x\}, 0 \le x \le 2\pi$ है। यदि उन बिन्दुओं की संख्या, जहाँ f अवकलनीय नहीं है, m है तथा उन बिन्दुओं की संख्या, जहाँ f संतत नहीं है, n है, तो क्रमित युग्म (m, n) बराबर है : **[JEE(Main)June2022]**

(A) (2, 0) (B) (1, 0) (C) (1, 1) (D) (2, 1)

Ans. B

Sol. $f(x) = \min\{1, 1 + x\sin x\}\ 0 \le x \le 2\pi$



$$f(x) = \begin{cases} 1 & 0 \le x < \pi \\ 1 + x\sin x & \pi \le x \le 2\pi \end{cases}$$

$f(0^+) = 1$ $f(0) = 1$

$f(\pi^-) = 1$ $f(\pi^+) = 1$ $f(\pi) = 1$

$f(2\pi^-) = 1$ $f(2\pi) = 1$

$n = 0$

$$f'(x) = \begin{cases} 0 & 0 < x < \pi \\ x\cos x + \sin x & \pi < x < 2\pi \end{cases}$$

$f'(\pi^-) = 0$ $f'(\pi^+) = \pi(-1) = -\pi$

$m = 1$

$(m, n) = (1, 0)$

18. Let $f : R \to R$ be defined as $f(x) = \begin{cases} [e^x], & x < 0 \\ ae^x + [x-1], & 0 \le x < 1 \\ b + [\sin(\pi x)], & 1 \le x < 2 \\ [e^{-x}] - c, & x \ge 2 \end{cases}$ where a, b, c $\in$ R and [t] denotes greatest integer less than or equal to t. Then, which of the following statements is true? **[JEE(Main)June2022]**

(A) There exists a, b, c $\in$ R such that f is continuous on R.

(B) If f is discontinuos at exactly one point, then $a + b + c = 1$

(C) If f is discontinuous at exactly one point, then $a + b + c \ne 1$

(D) f is discontinuous at atleast two points, for any values of a, b and c

माना $f: R \to R$ $f(x) = \begin{cases} [e^x], & x < 0 \\ ae^x + [x-1], & 0 \le x < 1 \\ b + [\sin(\pi x)], & 1 \le x < 2 \\ [e^{-x}] - c, & x \ge 2 \end{cases}$ द्वारा परिभाषित है, a, b, c ∈ R हैं तथा [t] महत्तम पूर्णांक t है। तो निम्न कथनों में से कौनसा सत्य है ? **[JEE(Main)June2022]**

(A) a, b, c ∈ R का अस्तित्व है जिनके लिए f, R पर संतत है

(B) यदि f मात्र एक बिन्दु पर असंतत है, तो a + b + c = 1

(C) यदि f मात्र एक बिन्दु पर असंतत है, तो a + b + c ≠ 1

(D) a, b, c के किन्हीं भी मानों के लिए f कम से कम बिन्दुओ पर असंतत है

Ans. C

Sol. f(x) is discontinuous at x = 1

For continuous at x = 0; a = 1

For continuous at x = 2; b + c = 1

a + b + c = 2

19. Let f, g : R → R be functions defined by $f(x) = \begin{cases} [x], & x < 0 \\ |1-x|, & x \ge 0 \end{cases}$ and $g(x) = \begin{cases} e^x - x, & x < 0 \\ (x-1)^2 - 1, & x \ge 0 \end{cases}$ where [x] denote the greatest integer less than or equal to x. Then, the function fog is discontinuous at exactly : **[JEE(Main)June2022]**

(A) one point (B) two points (C) three points (D) four points

माना फलन f, g : R → R $f(x) = \begin{cases} [x], & x < 0 \\ |1-x|, & x \ge 0 \end{cases}$ तथा $g(x) = \begin{cases} e^x - x, & x < 0 \\ (x-1)^2 - 1, & x \ge 0 \end{cases}$ द्वारा परिभाषित है, जहाँ [x] महत्तम पूर्णांक ≤ x है। तो फलन fog कितने बिन्दुओं पर असंतत है : **[JEE(Main)June2022]**

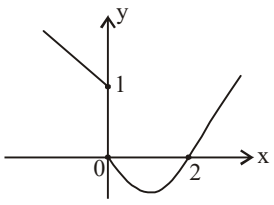
(A) एक बिन्दु (B) दो बिन्दु (C) तीन बिन्दु (D) चार बिन्दु

Ans. B

Sol. $f(x) = \begin{cases} [x] & ; \quad x < 0 \\ |1-x| & ; \quad x \ge 0 \end{cases}$ and $g(x) = \begin{cases} e^x - x & ; \quad x < 0 \\ (x-1)^2 - 1 & ; \quad x \ge 0 \end{cases}$

$$f(x)=\begin{cases}|1+x-e^x| & x<0\\ 1 & x=0\\ \left[(x-1)^2-1\right] & 0<x<2\\ \left|2-(x-1)^2\right| & x\geq 2\end{cases}$$

graph of fog(x)



So x = 0 and 2 are the 2 points where fog(x) is discontinuous.

20. The number of points where the function $f(x)=\begin{cases}|2x^2-3x-7| & \text{if } x\leq -1\\ [4x^2-1] & \text{if } -1<x<1\\ |x+1|+|x-2| & \text{if } x\geq 1,\end{cases}$

[t] denotes the greatest integer ≤ t, is discontinuous is ____. **[JEE(Main)June2022]**

उन बिन्दुओं, जहाँ फलन $f(x)=\begin{cases}|2x^2-3x-7| & \text{यदि } x\leq -1 \text{ है}\\ [4x^2-1] & \text{यदि } -1<x<1 \text{ है}\\ |x+1|+|x-2| & \text{यदि } x\geq 1 \text{ है,}\end{cases}$

[t] महत्तम पूर्णांक ≤ t है, असंतत है, की संख्या है _____ **[JEE(Main)June2022]**

Ans. 7

Sol. $\because$ f(–1) = 2 and f(1) = 3

For $x\in(-1,1)$, $(4x^2-1)\in[-1,3)$

hence f(x) will be discontinuous at x = 1 and also

whenever $4x^2-1=0, 1$ or 2

$\Rightarrow x=\pm\frac{1}{2},\pm\frac{1}{\sqrt{2}}$ and $\pm\frac{\sqrt{3}}{2}$

So there are total 7 points of discontinuity

21. Let f(x) = [2x² + 1] and $g(x)=\begin{cases}2x-3, & x<0\\ 2x+3, & x\geq 0\end{cases}$, where [t] is the greatest integer ≤ t. Then, in the open interval (–1, 1), the number of points where fog is discontinuous is equal to _____. **[JEE(Main)June2022]**

माना $f(x) = [2x^2 + 1]$ तथा $g(x) = \begin{cases} 2x-3, & x < 0 \\ 2x+3, & x \geq 0 \end{cases}$ हैं, जहाँ [t] महत्तम पूर्णांक $\leq t$ है। तो विवृत अंतराल (–1, 1) में उन बिंदुओं, जहाँ fog असंतत है, की संख्या है _____ **[JEE(Main)June2022]**

Ans. 62

Sol. $f(x) = [2x^2] + 1$

$$f(g(x)) = \left[2g^2(x)\right] + 1$$

$$f(g(x)) = \begin{cases} \left[2(2x-3)^2\right]+1 & , \quad x \in (-1,0) \\ \left[2(2x+3)^2\right]+1 & , \quad x \in [0,1) \end{cases}$$

(i) $x \in (-1, 0)$

$2x \in (-2, 0)$

$2x - 3 \in (-5, -3)$

$(2x - 3)^2 \in (9, 25)$

$2(2x - 3)^2 \in (18, 50)$

f(g(x)) is DC when

$2(2x - 3)^2 \in \{19, 20, ...,49\}$

Total = 31

(ii) $x \in [0, 1)$

$2x \in [0, 2)$

$2x + 3 \in [3, 5)$

$(2x + 3)^2 \in [9, 25)$

$2(2x - 3)^2 \in [18, 50)$

f(g(x)) is DC when

$2(2x + 3)^2 \in \{19, 20, ...,49\}$

Total = 31

Total point of discontinuity = 31 + 31 = 62

22. If the function $f(x) = \begin{cases} \dfrac{\log_e(1-x+x^2)+\log_e(1+x+x^2)}{\sec x - \cos x} & , \quad x \in \left(\dfrac{-\pi}{2}, \dfrac{\pi}{2}\right) - \{0\} \\ k & , \quad x = 0 \end{cases}$ is continuous at

x = 0, then k is equal to : **[JEE(Main)July2022]**

यदि फलन $f(x) = \begin{cases} \dfrac{\log_e(1-x+x^2)+\log_e(1+x+x^2)}{\sec x - \cos x} & , \quad x \in \left(\dfrac{-\pi}{2}, \dfrac{\pi}{2}\right) - \{0\} \\ k & , \quad x = 0 \end{cases}$

x = 0, पर संतत है, तो k बराबर है : **[JEE(Main)July2022]**

(A) 1 (B) –1 (C) e (D) 0

Ans. A

Sol. $\lim\limits_{x\to 0}\dfrac{\left(\ln\left(1+x^2+x^4\right)\right)\cos x}{1-\cos^2 x}$

$$\lim_{x\to 0}\frac{\left(\dfrac{\ln\left(1+x^2+x^4\right)}{x^2+x^4}\right)x^2\left(1+x^2\right)\cos x}{\left(\dfrac{\sin^2 x}{x^2}\right)x^2}=1$$

$\therefore$ k = 1

23. If $f(x)=\begin{cases}x+a & , \ x\le 0\\ |x-4| & , \ x>0\end{cases}$ and $g(x)=\begin{cases}x+1 & , \ x<0\\ (x-4)^2+b & , \ x\ge 0\end{cases}$ are continuous on R, then (gof) (2) + (fog) (–2) is equal to : **[JEE(Main)July2022]**

यदि $f(x)=\begin{cases}x+a & , \ x\le 0\\ |x-4| & , \ x>0\end{cases}$ तथा $g(x)=\begin{cases}x+1 & , \ x<0\\ (x-4)^2+b & , \ x\ge 0\end{cases}$ R पर संतत तो (gof) (2) + (fog) (–2) बराबर है: **[JEE(Main)July2022]**

(A) –10 (B) 10 (C) 8 (D) –8

Ans. D

Sol. $f(x)=\begin{cases}x+a, & x\le 0\\ |x-4|, & x>0\end{cases}$ and $g(x)=\begin{cases}x+1, & x<0\\ (x-4)^2+b, & x\ge 0\end{cases}$

$\because$ f(x) and g(x) are continuous on R

$\therefore$ a = 4 and b = 1 – 16 = –15

then (gof)(2) + (fog) (–2)

= g(2) + f(–1)

= –11 + 3 = – 8

24. If for $p\ne q\ne 0$, the function $f(x)=\dfrac{\sqrt[7]{p(729+x)}-3}{\sqrt[3]{729+qx}-9}$ is continuous at x = 0, then : **[JEE(Main)July2022]**

यदि $p\ne q\ne 0$ के लिए, फलन $f(x)=\dfrac{\sqrt[7]{p(729+x)}-3}{\sqrt[3]{729+qx}-9}$, x = 0 पर संतत है, तो : **[JEE(Main)July2022]**

(A) 7pq f(0) – 1 = 0 (B) 63q f(0) – p^2 = 0

(C) 21q f(0) – p^2 = 0 (D) 7pq f(0) – 9 = 0

Ans. B

Sol. $f(0)=\lim_{x\to 0} f(x)$

Limit should be $\frac{0}{0}$ form

So, $\sqrt[7]{p.729}-3=0 \Rightarrow p.3^6=3^7 \Rightarrow p=3$

Now, $f(0)=\lim_{x\to 0}\frac{\sqrt[7]{3\left(3^6+x\right)}-3}{\sqrt[3]{3^6+qx}-9}$

$$=\lim_{x\to 0}\frac{3\left[\left(1+\frac{x}{3^6}\right)^{1/7}-1\right]}{9\left[\left(1+\frac{qx}{3^6}\right)^{1/3}-1\right]}=\frac{3}{9}\times\frac{\frac{1}{7.3^6}}{\frac{q}{3.3^6}}$$

$$\Rightarrow f(0)=\frac{1}{3}\times\frac{3}{7q}=\frac{1}{7q}$$

$\Rightarrow 7qf(0)-1=0$

$\Rightarrow 7.p^2.qf(0)-p^2=0$ (for option)

$\Rightarrow 63qf(0)-p^2=0$

25. The function $f: R \to R$ defined by $f(x)=\lim_{n\to\infty}\frac{\cos(2\pi x)-x^{2n}\sin(x-1)}{1+x^{2n+1}-x^{2n}}$ is continuous for all x in :

[JEE(Main)July2022]

$f(x)=\lim_{n\to\infty}\frac{\cos(2\pi x)-x^{2n}\sin(x-1)}{1+x^{2n+1}-x^{2n}}$ द्वारा परिभाषित फलन $f: R \to R$ किस समुच्चय के सभी बिन्दु x पर संतत है:

[JEE(Main)July2022]

(A) R – {–1} (B) R – {–1, 1} (C) R – {1} (D) R – {0}

Ans. B

Sol. n should be given as a naturel number.

$$f(x=\begin{cases}\frac{-\sin(x-1)}{x-1} & x<-1\\ -(\sin 2+1) & x=-1\\ \cos 2\pi x & -1<x<1\\ \frac{-\sin(x-1)}{x-1} & x>1\end{cases}$$

f(x) is discontinuous at x = – 1 and x = 1

26. The number of points, where the function $f: R \to R$,

$f(x) = |x-1| \cos|x-2| \sin|x-1| + (x-3)|x^2-5x+4|$, is NOT differentiable, is :

[JEE(Main)July2022]

उन बिन्दुओं की संख्या, जहाँ पर फलन $f: R \to R$,

$f(x) = |x-1| \cos|x-2| \sin|x-1| + (x-3)|x^2-5x+4|$, अवकलनीय नहीं है, है : **[JEE(Main)July2022]**

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $f(x) = |x-1| \cos|x-2| \sin|x-1| + (x-3)|x^2-5x+4|$

$= |x-1| \cos|x-2| \sin|x-1| + (x-3)|x-1||x-4|$

$= |x-1| [\cos|x-2| \sin|x-1| + (x-3)|x-4|]$

Non differentiable at $x = 1$ and $x = 4$.

27. Let the function $f(x) = \begin{cases} \dfrac{\log_e(1+5x) - \log_e(1+\alpha x)}{x} & ; \text{if} \quad x \neq 0 \\ 10 & ; \text{if} \quad x = 0 \end{cases}$ be continuous at $x = 0$. Then α is equal to :

[JEE(Main)July2022]

माना फलन $f(x) = \begin{cases} \dfrac{\log_e(1+5x) - \log_e(1+\alpha x)}{x} & ; \text{if} \quad x \neq 0 \\ 10 & ; \text{if} \quad x = 0 \end{cases}$, $x = 0$ पर संतत है। तो α बराबर है :

[JEE(Main)July2022]

(A) 10 (B) –10 (C) 5 (D) –5

Ans. D

Sol. $f(x) = \begin{cases} \dfrac{\ln(1+5x) - \ln(1+\alpha x)}{x} & ; x \neq 0 \\ 10 & ; x = 0 \end{cases}$

$\lim\limits_{x \to 0} \dfrac{\ln(1+5x) - \ln(1+\alpha x)}{x} = 10$

Using expension

$\lim\limits_{x \to 0} \dfrac{(5x + \ldots\ldots) - (\alpha x + \ldots..)}{x} = 10$

$5 - \alpha = 10 \Rightarrow \alpha = -5$

28. $f(x)=\begin{cases}\left|4x^2-8x+5\right|, & \text{if} \quad 8x^2-6x+1\geq 0\\ \left[4x^2-8x+5\right] & \text{if} \quad 8x^2-6x+1<0\end{cases}$ where $[\alpha]$ denotes the greatest integer less than or equal to α.

Then the number of points in R where f is not differentiable is _____. **[JEE(Main)July2022]**

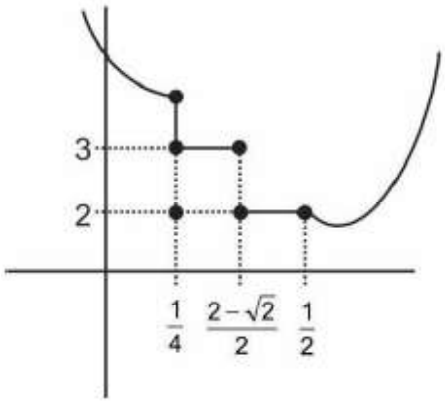
माना $f(x)=\begin{cases}\left|4x^2-8x+5\right|, & \text{if} \quad 8x^2-6x+1\geq 0\\ \left[4x^2-8x+5\right] & \text{if} \quad 8x^2-6x+1<0\end{cases}$ जहाँ $[\alpha]$ महत्तम पूर्णांक α है। तो R में उन बिन्दुओं की संख्या, जहाँ f अवकलनीय नहीं है, है _____ **[JEE(Main)July2022]**

Ans. 3

Sol. $f(x)=\begin{cases}\left|4x^2-8x+5\right|, & \text{if } 8x^2-6x+1\geq 0\\ \left[4x^2-8x+5\right], & \text{if } 8x^2-6x+1<0\end{cases}$

$$=\begin{cases}4x^2-8x+5, & \text{if } x\in\left[-\infty,\frac{1}{4}\right]\cup\left[\frac{1}{2},\infty\right)\\ \left[4x^2-8x+5\right] & \text{if } x\in\left(\frac{1}{4},\frac{1}{2}\right)\end{cases}$$

$$f(x)=\begin{cases}4x^2-8x+5 & \text{if } x\in\left(-\infty\frac{1}{4}\right]\cup\left[\frac{1}{2},\infty\right)\\ 3 & x\in\left(\frac{1}{4},\frac{2-\sqrt{2}}{2}\right)\\ 2 & x\in\left[\frac{2-\sqrt{2}}{2},\frac{1}{2}\right)\end{cases}$$



$\therefore$ Non–diff at $x=\frac{1}{4},\frac{2-\sqrt{2}}{2},\frac{1}{2}$

29. If [t] denotes the greatest integer ≤ t, then the number of points, at which the function $f(x)=4|2x+3|+9\left[x+\frac{1}{2}\right]-12[x+20]$ is not differentiable in the open interval (–20, 20), is _____.

[JEE(Main)July2022]

यदि [t] महत्तम पूर्णांक ≤ t है, तो उन बिन्दुओं, जिन पर फलन $f(x)=4|2x+3|+9\left[x+\frac{1}{2}\right]-12[x+20]$ विवृत अंतराल (–20, 20) में अवकलनीय नहीं है, कि संख्या_____ **[JEE(Main)July2022]**

Ans. 79

Sol. $f(x)=4|2x+3|+9\left[x+\frac{1}{2}\right]-12[x+20]$

$x\in(-20,20)$

f(x) is not Diff. at $x=I\in\{-19,-18,\ldots\ldots 19\}=39$

at $x=I+\frac{1}{2}$, $f(x)$ Not diff. at 39 points

Check at $x=\frac{-3}{2}$ Discount at $x=\frac{-3}{2}\therefore$ N.R(1)

No. of point of non-differentiabilty

= 39 + 39 + 1 = 79